

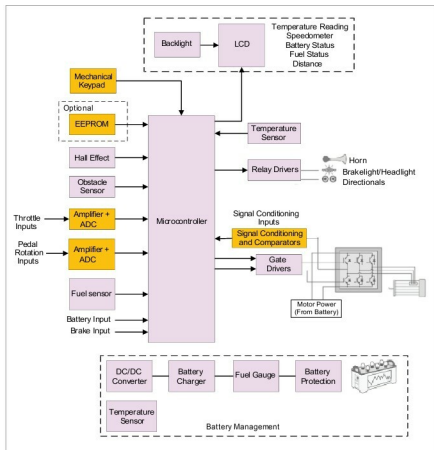


Fig. 2: Microcontrollers in automotive e-bike applications

tor. The MCU receives vehicle input signal and the vehicle starts moving. The MCU uses driver circuitry to drive the three-phase brushless automotive motor as per speed required by the user. Speed of the motor varies and is controlled as per acceleration brake sensor input from the user.

The MCU uses either internal or external serial EEPROM (I2C/SPI based) for storing data like distance readings. It uses RTC for displaying accurate time on the display. Temperature measurement is done by using an onboard RTD or thermistor based temperature-sensing device.

The e-bike solution in the automotive industry uses an obstacle sensor to get information about nearby vehicles while parking, a fuel sensor to get information regarding the amount of fuel in the tank, while an MCU monitors battery voltage and shows it on the LCD

display. The MCU uses relay-driver circuitry for switching brakelight/headlight on or off and for blinking directional lights.

The power supply section uses rechargeable lead-acid/lithium battery as the power source. It also has provision for a battery charger. Battery input is down-converted to DC voltage to power the MCU and other circuitry. Ignition key of the e-bike enables and disables onboard regulators.

The power supply section incorporates protection features for battery, over-current, over-heating and start-up fail, which are controlled by the MCU. It also enables charging of external devices like mobile phones.

Application in mobile phones and tablets

Current mobile phone and tablet designs use 8/16/32-bit MCUs as

TRMS DIGITAL MULTIMETERS

153B+ TRMS

171B+ TRMS

3% Digit 6000 Count LCD with Backlight

- Auto Ranging
- 600mV ~ 1000V AC TRMS
- 600mV ~ 750V AC TRMS
- 600µA ~ 20A DC
- 600µA ~ 20A AC TRMS
- 600Ω ~ 60MΩ
- 9.999nF ~ 9.999µF
- 99.99Hz ~ 20.00MHz
- 0.1% ~ 99.9% Duty Cycle
- -40°C ~ 1000°C (-40°F ~ 1832°F) (171B+ TRMS)
- K Type Thermocouple (Upto 260°C) (171B+ TRMS)
- REL Δ, Max/Min (171B+ TRMS)
- Diode Test, Data Hold, Audible Continuity, Auto Power off & Low Battery Indicator

Meco Meters Private Limited

Plot No. EL-60, MIDC Electronic Zone,
TTC Industrial Area, Mahape,
Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-27673300 (Board), 27673311-16 (Sales)
Fax : 0091-22-27673310, 27673330
Email : sales@mecoinst.com
Web : www.mecoinst.com